Appendix: China's infrastructure finance in Africa

Country Perspectives

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Please not references for *Appendix: Country Perspectives* can be found in the working paper, *China's infrastructure finance in Africa: impacts and lessons* (www.odi.org/en/publications/chinas-infrastructure-finance-in-africa-impacts-and-lessons/).

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Hydropower

Ethiopia

Benefits: China has been an important partner in Ethiopia's power sector development, providing almost $800 million in financing for the three plants comprising the Gilgel Gibe III (1,870MW) megaproject, construction of which began in 2006 and was completed in 2015, as well as the smaller Genale Dawa III (250MW) and Finchaa-Amerti-Neshe (97MW) dams. China has financed and constructed[1] transmission lines designed to connect Ethiopia's largest hydropower project – Grand Renaissance – to the national grid. As a result, Chinese-financed hydropower plants in Ethiopia account for 20% of overall installed generation capacity in the country. In the two years following the commissioning of Gilgel Gibe III in 2016, electricity generation in Ethiopia surged by 30%. Chinese contractors have also supplied technical skills.

Challenges: The construction phase for Gilgel Gibe III suffered time and cost over-runs. The official environmental and social impact assessment was delayed for two years after the start of the project. Due to technical limitations of the counterpart national utility, EEPCO, it was challenging to integrate Gilgel Gibe III into the Ethiopian grid. Lack of planning and coordination by EEPCO meant that delays in the completion of related transmission projects needed to evacuate power from Gilgel Gibe III prevented the supply of electricity from the project becoming fully available to consumers. In addition, due to the exceptionally long transmission line connecting Gilgel Gibe III to the grid, there have been technical issues in managing voltage stability and compensation for the new hydropower project.

Since generation and transmission were contracted to separate construction firms, neither had responsibility for ensuring compatibility with the grid. In addition, the price of electricity negotiated under the Power Purchase Agreements for Gilgel Gibe III significantly exceeded the retail price of electricity in Ethiopia, meaning that every kilowatt-hour (kWh) of electricity supplied by Gilgel Gibe III would generate a financial loss for EEPCO. These problems reflect the fact that the economic and financial analyses were more cursory than the technical feasibility studies. They also reflect the fact that requisite sector reforms were not/could not be done.

Ghana

Benefits: Plans for the Bui dam date to the 1960s, based on technical support received from Australia and the World Bank. However, it was only in the 1990s that a feasibility study was conducted by Coyne et Bellier (now Tractebel), together with an environmental assessment by the University of Aberdeen (Power Technology, 2014). In 1991, the Volta River Authority authorised Halliburton and Brown and Root to construct the dam, but in 2001 this process was stalled by the Government of Ghana. The project finally became a reality when Sinohydro submitted a bid with funding from China Eximbank in 2005. The Ghanaian Government signed an EPC contract with Sinohydro Corporation. In 2007, Ghana obtained a buyer's credit loan of $293.5 million with a grace period of 5 years with a 12-year maturity period, and an interest rate set at a margin of 1.075% over the prevailing commercial interest reference rate (Gocking, 2020). This was followed by a concessional

[1] The transmission line spans 620km, and will carry 600MW, 500kV 4 circuit transmission.

loan for 2.1 billion renminbi (RMB) ($330 million) at 2% fixed interest, a 5-year grace period, and a maturity of 20 years, both from China Eximbank. The granting of the loan was followed by an environmental impact assessment and a feasibility study. The creation of the Bui Power Authority in August 2007 accelerated the building of the dam, which was commissioned in 2013, boosting the country's installed capacity by 404MW. The project also included the construction of 241km of transmission lines and resettlement of a village (Bui Power Authority, 2024).

Challenges: The non-concessional export credit was secured with revenue from the sale of the dam's power, as well as an agreement to sell 30,000 metric tons of cocoa from Ghana, annually, at the prevailing market price through a cocoa sales agreement, the latter being partly motivated by a desire to promote cocoa consumption in China (Hensengerth, 2011). The proceeds from the cocoa sales were to be put into an escrow account to help repay the debt. In addition, to ensure that the loan would be fully repaid, the terms required that electricity from the Bui dam sell for not less than $0.03–0.04 per kWh. The project eventually required a further $168 million and additional rescheduling. This was attributed to unanticipated effects of the 2008 financial crisis, unforeseen additional work and underestimation of the EPC costs (All Africa, 2011). The commissioning of the dam in 2013 coincided with a period of drought, and it had no immediate impact on the volume of electricity generation.

Guinea

Benefits: Despite having vast hydropower resources, Guinea has long struggled to develop its electricity sector due to the scale of the investments involved and financing required. To remedy this, a partnership with China's CWE was established in 2011 for the construction of the 234MW Kaleta hydropower plant, commissioned in 2015. The success of this project, together with continued rapid growth in electricity demand, led to a further agreement with CWE in 2018 for a second and larger hydropower project, Souapiti (515MW) – dubbed the 'Three Gorges' of Guinea – which was commissioned in 2021. Chinese financing for construction is estimated at over $1.5 billion via a preferential buyer's credit from China Eximbank, with a further $0.5 billion contribution from the government, at a 2% interest rate, 7-year grace period and 20-year maturity.[2] Loan proceeds were then on-lent to a special purpose vehicle (SPV) jointly owned by the Guinean Government and CWE. The government sold a 51% stake in Kaleta to CWE as a means of raising finance for Souapiti, in which the company also holds a 49% stake and acts as the EPC contractor.

The commissioning of the first smaller plant, Kaleta, led to an immediate surge in power supply equivalent to a 40% expansion in the country's electricity generation in 2015, followed by a further 11% increase in 2016. This increase in power generation provided immediate relief from power shortages, stabilising power supply and allowing the phaseout of emergency rental plants. Power outages reported by industrial customers declined from 214 hours per month in 2006 to 14 hours per month in 2016 (World Bank Enterprise Surveys, 2016). The completion of Souapiti in 2021 doubled Guinea's hydroelectric production capacity, creating a potential surplus of

[2] Based on AidData project data. See: https://china.aiddata.org/projects/187/

energy for export to the West Africa Power Pool, contingent on the completion of the associated regional interconnectors.

Challenges: The Souapiti project in Guinea faced significant implementation challenges. The associated resettlement of people, managed by the local authorities, was the largest in Guinea's post-independence history. The management of this process created controversy regarding the associated social and environmental impacts of the project and could have been better handled. Planning and coordination issues also arose, with delays in the required parallel expansion of transmission lines from Souapiti and Kaleta to Conakry, which left large amounts of new generation power stranded. Drought conditions also subsequently affected power generation from Souapiti. In addition, the scale of Souapiti relative to the size of the Guinean economy made the project macroeconomically challenging. For instance, foreign direct investment surged to 16% of GDP in 2018 due to the construction of Souapiti, inflating the current account deficit, while public debt rose sharply by 7 percentage points of GDP. The price agreed in the Power Purchase Agreements for the two hydropower plants with the national utility, EDG, averages prices across the two projects at $0.1075. This is a fraction of what the country was previously paying for emergency thermal generation, but well above the electricity retail price, necessitating substantial subsidies from the Ministry of Finance.

Zambia

Benefits: Zambia has suffered from chronic electricity shortages and underinvestment in the power sector. Chinese engagement has focused on two major hydropower generation projects – Lake Kariba North (360MW) and Kafue Gorge Lower (750MW) – agreed around 2009/10. The development of these two projects was supported by $1.5 billion in Chinese lending from two parties, China Eximbank and ICBC. Sinohydro constructed both dams and stations. The two projects have made a material contribution to Zambia's power system, representing about 30% of the entire national installed generation capacity of 3–4GW.

Challenges: The onset of drought conditions has critically impacted the generation capacity of these dams, particularly Lake Kariba, leading to blackouts across the country and the declaration of a state of emergency in March 2024 (Euronews, 2024). Furthermore, the financial contracts of the two major Chinese-financed hydropower projects were problematic. Electricity generated by the Kafue Lower Gorge dam was contracted to be supplied at $0.13–0.16 per kilowatt-hour under the Power Purchase Agreement, eventually dropping to $0.08–0.10 per kilowatt-hour once the loan to China was repaid. Given that electricity retail tariffs in Zambia were then only around $0.05–0.06 per kilowatt-hour, this electricity was to be sold at a loss.

Rail

Angola

In Angola, Chinese infrastructure finance has been supported through a long-term financing agreement, based on a high-level inter-governmental negotiation, and oriented towards agreed projects aligned with national development plans.

Following the end of the civil war in 2002, Angola became one of China's main trading partners on the continent, primarily of petroleum resources, as well as the largest bilateral recipient of loan commitments. Much of this was channelled via large framework agreements, beginning in 2004, when China Eximbank approved a $2 billion oil-backed loan, with a 3-year grace period and 21-year tenor at a LIBOR 1.6 basis points (bps) margin. This was secured with an oil prepayment facility typical of China's pre-export finance structures in this period (Lui and Chen, 2021), eventually labelled the 'Angola model'.[3]

Proceeds from the loan were assigned to individual projects – which included the major rehabilitation of the Benguela railway – at the instruction of Angola's Ministry of Finance, with payments to contractors made by China Eximbank (Corkin et al., 2008). In turn, payments for oil imports were deposited in an escrow account that serviced the outstanding debt as part of a ringfencing arrangement, reducing the risk to the creditor. Subsequent agreements with Eximbank were signed in 2006 (another $2 billion facility) and several agreements with CDB, including a large bilateral agreement with CDB and Sinosure in 2016.

Benefits: Angola's rail infrastructure was largely destroyed during the civil war (China Lusophone Brief, 2018). At the end of the conflict, the new President created the Office of National Reconstruction to ensure that investments important for reconstruction were delivered effectively and to have direct control/discretionary authority over them (Wang, 2023). The country's most strategic rail line (and a very profitable one before the war), the *Caminho de Ferro de Benguela* (CFB), known as the Benguela railway, links the Atlantic port of Lobito to the land border with the DRC. There it connects with the Congolese rail network operated by Société Nationale de Chemins de Fer du Congo (SNCC), providing the shortest route to the sea for the rich copper mining belt in DRC and Zambia. After the war, the new government reached out to Western donors but, dissatisfied with their conditions, turned to China. A Chinese loan financed the rehabilitation of the 1,344-kilometre CFB line, including 67 stations, a process that was completed in 7 years. The line was also upgraded to allow for faster trains (90km/hr) and rising traffic volumes. The work was carried out entirely by Chinese companies, and also involved supply of rolling stock and subsequent maintenance and operations. Contractual terms restricted the tendering process for infrastructure projects under the loan agreement, requiring 70% of the total value of all contracts to go to Chinese companies with the remainder available for Angolan firms (Haroz, 2011). In 2023, the Lobito

[3] This means that repayments to Eximbank were made from revenues of oil sales collected under a sales contract between Sonangol and UNIPEC (China international United Petroleum & Chemicals Co. Ltd: Sinopec group), according to market prices.

Atlantic Railway Company (non-Chinese) was awarded a concession to build additional lines and provide railway services (Lobito Corridor Investment Promotion Agency, 2024).

Challenges: The construction of the Benguela Railway was managed first by the ONR and the China International Fund (CIF) (Wang, 2023),[4] then by the Ministry of Transport. The project suffered from financial and managerial misgovernance due to deficient oversight arrangements. It also became clear that lack of technical capacity as well as poor communication between Chinese contractors and Angolan civil servants would hinder completion, leading to delays in construction and two instances where construction was halted.

Furthermore, the anticipated flow of mineral traffic from Central Africa, which had been estimated to be as high as 300,000–1,000,000 tons per year, failed to materialise. The design of the line was not informed by the technical requirements of the mining companies it was intended to serve. A post-construction assessment concluded that CFB did not meet the technical standards for use as the main route for export of ore from the Copper Belt in the Southern African hinterland (China Lusophone Brief, 2018). A proper feasibility study for the project was also lacking. Redressing these issues required substantial investment estimated at $250 million for infrastructure upgrades and $150 million for acquisition of rolling stock, as well as improvements to maintenance and traffic management systems (MINTRANS, 2019).

Given lower than anticipated cross-border mineral freight, CFB operates at a fraction of its design capacity, carrying mainly passengers paying nominal fares. Freight services are infrequent and only run on demand, with traffic consisting primarily of fuel and other bulk commodities. Copper and manganese, the main anticipated traffic on the line, currently account for less than 20% of total volumes (Raina et al., 2022). As a result, rail revenues barely cover staff salaries, and CFB depends on a state subsidy for 50% of revenues (Ibid.). Service of the debt associated with the CFB line falls to the government, which is already highly indebted.

Ethiopia

In Ethiopia, China Eximbank has supported infrastructure finance via major framework agreements in 2009, with a pledge to support 15 infrastructure projects totalling $7.6 billion over 2011 to 2017, including the Addis–Djibouti standard gauge railway (SGR). These projects formed part of the country's Growth and Transformation Plans (GTPs), which sought to leverage Chinese capital, but also draw from the East Asian development experience through investment in infrastructure and industrialisation to support an export-led growth model. This bilateral agreement secured the entry of Chinese contractors and suppliers into the market and helped enable the construction of major infrastructure investments under national development plans.

Benefits: In 2007, the Ethiopian Railway Corporation (ERC) was created to oversee the construction of a new and fully electrified national rail network spanning 5,000km. This network was to support an industrialisation and export-oriented growth strategy by connecting major planned industrial

[4] The CIF was established around 2003 by a group of Chinese individuals in Hong Kong with the goal of infrastructure investment.

zones across the country – including many with Chinese involvement – to the seaport in Djibouti (also financed and constructed by Chinese financial institutions and SOEs). The 756km Addis Ababa–Djibouti railway cost a total of $4.3 billion, of which $2.6 billion was an export credit from China Eximbank (Chen, 2021). The Governments of Ethiopia and Djibouti provided the balance, with Ethiopia guaranteeing Djibouti's contribution for construction of the line (Dreher et al., 2017). The bulk of the Chinese finance provided to ERC was contracted at a variable Libor 6-month plus 3.0% interest rate and initially had a 20-year maturity (15 years plus a 6-year grace period) with repayment starting in 2017.[5]

Construction began in 2011 and lasted five years. Under the terms of the loans, contractors were selected through a competitive process confined to Chinese enterprises, and an EPC contract was awarded to a consortium comprising CREC and CCECC. Contractors also undertook to follow local environmental and social safeguards during the construction. This primarily entailed maintaining the original landscape along the railway and constructing wildlife crossings.

Training and technology transfer have been a major feature of the project. To operate the line, a joint venture, the Ethiopia–Djibouti Railway (EDR) was formed, with Ethiopia having a 75% share and Djibouti 25%. Due to a lack of skilled local personnel, CREC and CCECC, which were originally contracted under an EPC, took on an operations and management (O&M) contract (costing $417 million), to stay on and manage railway operations for a six-year period before handing over to local operators by 2024 (myNEWS, 2024). Under the agreement, Chinese contractors undertook to send ERC employees to technical universities in China and established training centres on-site to train drivers, engineers and manual workers (Chen, 2024).

Challenges: Numerous factors have stymied operations, including sector policies, technical design issues, conflicting regulations and poor management of overland freight traffic. Revenue forecasts were a third of what had been envisaged in the original feasibility study. To break even, including full costs of debt service, EDR needed to capture a little over half the total cargo volume between Addis Ababa and Djibouti, which amounted to 10 million tonnes. This required a substantial modal shift, as rail captured just 5% of total cargo traffic on this route. Most freight went by road, despite road freight tariffs costing 30% more than rail, and road journeys taking longer.

Freight traffic in Ethiopia is highly regulated, with both prices and modal choices controlled by the Ethiopian Shipping Logistics Services Enterprise (ESLSE), which operates its own trucking fleet, and which viewed the railway as a direct competitor. Industrial zones outside the corridor would have relied on trucking shipment to the railway dry port at an additional cost. In 2023, the government partially liberalised the logistics sector by providing licences to a few private players, potentially breaking the pricing bottleneck (Africa Intelligence, 2023).

The railway's operations were further hampered by lack of institutional capacity, power supply problems and overall poor planning. These factors resulted from a combination of low capacity, poor governance and tight fiscal

[5] This was estimated to be around 5.15% prior to Covid-19. The loan tenor was extended from 20 to 30 years in 2018 following a bilaterally negotiated debt restructuring.

circumstances. The rail line was constructed as an electric line in a country where the energy system was not ready to support it.

Despite timely completion in 2016, delays in the construction of the associated electricity transmission network held up the commissioning of the railway until 2018. Even then, poor energy sector planning, shortages of generation capacity and lack of necessary transmission and distribution infrastructure led to ongoing technical problems in the form of power outages and voltage surges, leading to frequent suspension of services during early operation. The absence of direct rail links into industrial zones has also affected the railway's ability to capture more traffic from export-oriented firms.

The EDR has also been impacted by wider conflicts, including tensions regarding overcompensation for rural roads on which the rail lines were built, vandalism to rail infrastructure and tension with local communities concerning animal casualties on the tracks (The Economist, 2018).

In the face of these difficulties and a deteriorating budget deficit, in 2018 Ethiopia renegotiated the tenor of the loan to 30 years. The Chinese contractors continued to fulfil the operation and management contract, despite arrears on debt owed by EDR. Reports in 2021 showed an improvement in revenues of over one-third from 2020 due to higher volumes of imports, at $86 million. The railway line broke even in 2021 (IRJ, 2022). However, the line's operations have been hampered by the civil war in Tigray from 2020 to 2022, and railway loans have contributed to Ethiopia's debt problems. It has applied to the G20 Common Framework to restructure its official and commercial borrowing.

Kenya

Benefits: The construction of Kenya's standard guage railway (SGR) was originally intended to be the first stage of a larger regional infrastructure programme of the East African Community (EAC) (CPCS, 2009). In 2004, EAC members launched an EAC railway master plan to improve rail links. The plan was completed in 2009. Kenya's Vision 2030 envisaged a large investment in the Port of Mombasa and rail freight consistent with its growth plan. Kenya is a core partner in the Belt and Road Initiative (BRI), with its involvement centred around the SGR linking Mombasa to Nairobi. The China Road and Bridge Corporation (CRBC), a large state-owned engineering company, proposed an Engineering, Procurement and Construction (EPC) arrangement to the government to build the railway, with the added attraction of potential additional Chinese financing. The CRBC had built a section of the Shanghai–Beijing high speed rail project and had experience with large projects.

In 2009, CRBC was asked to carry out a feasibility study and a preliminary design for the SGR (Brautigam et al., 2022). The study concluded that the project had a high rate of return; CRBC was engaged to construct the Mombasa–Nairobi portion with financing from China Eximbank. The project was funded with a $1.6billion subsidised export credit at 2% interest (grace period 7 years and repayment 13 years) and a commercial loan of $2 billion (repayment 10 years, grace period 5 years).

Several measures were established to provide credit enhancement for the loan. These included use of a railway development levy to repay the loan, a policy directive to transfer road freight to rail, and an off-take arrangement with the Kenya Port Authority which uses an escrow account funded by revenues as security for the Eximbank loan (Brautigam, Bhalaki and Deron, 2022).

The 472km Mombasa–Nairobi rail line was completed in 2017. Construction proceeded swiftly under the leadership of the CRBC, which completed the line in 32 months, well under the 60 months allowed by the contract. The line became operational in 2017 and has seen significant increases in traffic revenues, reaching $126 million in 2018.

Subsequently, China also funded the Nairobi–Naivasha portion, which measures 120 kilometres and had an estimated cost of $1.5 billion. More recently, Kenya has announced plans for funding for the remaining segment from Naivasha to Malaba, this time under a PPP structure, leveraging funding from private Chinese investors and commercial sources, with Eximbank financing playing a minor role (Nyabiage, 2025).

Challenges: The implementation of Kenya's SGR project raised a range of major challenges (Taylor, 2020). To accelerate construction and implementation of the SGR, China and Kenya utilised a government-to-government procurement model, which exempted the project from Kenya's Public Procurement and Disposal Act. The project was not subject to competitive bidding or reporting requirements for government tenders, but was handled as a single-source procurement. This prevented a market test for the estimated costs of the project. Construction costs were substantially higher than expected, working out at around $5 million per kilometre of line on average (Carrai, 2021). Subsequently, there were repeated allegations of corruption in connection with the SGR and public concern over inflated costs (Transport, Public Works and Housing, 2020). Investigations were launched by Kenya's Public Prosecutor in 2018 and the Ethics and Anti-Corruption Commission in 2019.

Revenues earned by the new Mombasa–Nairobi line fell short of the operating costs of $170 million; the difference was partially covered by a government subsidy of $49 million in 2018 (Business Daily, 2020). The subsidy did not ensure that all payments were made on time. Operating costs of the rail line included the management fee, of around $1 million a month to Afristar, a subsidiary company of CRBC, which took on contractual responsibility for operating the line until 2024. However, due to chronic default on the payment of the management fee, the management contract was cancelled in 2021 and operations reverted to the national rail company, KRC (IRJ, 2021).

Although the debt for the railway construction was transferred to the books of KRC, the loss-making nature of the line means that there is no possibility of servicing the associated Chinese loans without recourse to government resources. The conclusion of the grace periods on Kenya's rail-related Chinese loans in 2019 and 2021, respectively, has placed Kenya in a difficult financial position. It has annual dollar-dominated debt service of $800 million partially pegged to rising interest rates (Chatham House, 2023). In view of these challenges, China withheld further funding that would have gone towards extending the SGR line from central Kenya to the border with Uganda. China Eximbank asked Kenya to redo a feasibility study to prove commercial viability before extending financing.

ICT and digital technologies

The examples below provide further insights into the structuring of Chinese ICT engagement in two of the most salient countries – Ethiopia and Kenya – highlighting both development benefits and implementation challenges.

Ethiopia

Benefits: Ethiopia is the main recipient of Chinese infrastructure finance for ICT. In 2006, the Ethiopian Telecommunication Corporation (ETC) signed what was at the time the largest agreement in the history of telecommunications in Africa. The 'Millennium Plan' project provided a loan of $1.5 billion (to which ZTE added $400 million for engineering services) to overhaul and expand Ethiopia's telecommunications system. The project was divided into three phases that covered different parts of the country's network. The first phase of the project connected the ICT network of Ethiopia's 13 largest cities, while the second and third phases expanded coverage to rural areas (Gagliardone, 2016). The second phase focused on the Global System for Mobile (GSM) Communications and Internet Protocol (IP) based systems that cover cities and roads. The third phase extended further out to cover rural and remote areas. The project achieved notable results: internet and data subscribers grew from 71,059 in 2009 to around 221,000 by the end of 2012 (Custer et al., 2023). While the project did not fund mobile devices, the number of mobile subscribers also grew from less than 1.2 million in September 2007 to around 17.5 million in 2012, possibly partly due to increased internet access.

Following the Millennium Project, in 2013 Huawei and ZTE were tasked with the implementation of the Telecom Transformation and Expansion Circles Project to provide fourth generation (4G) technology, a broadband technology allowing browsing speeds of 100mb per second, in Addis Ababa, and to enable third generation (3G) services across the country.

Overall, there were important improvements in terms of network coverage and access. For instance, internet users have grown from less than 1% of the population in 2001 to nearly 20% in 2022 (International Telecommunication Union, n.d.). While this growth cannot be exclusively attributed to Chinese financing for ICT, Chinese financed projects have certainly contributed.

Challenges: Chinese lending was to the Ethiopian state monopoly (Ethiopian Telecommunications Corporation (ETC), later renamed Ethio Telecom) supported by ZTE. Monopolisation of the sector meant that there was little incentive to keep costs and prices low and quality/access high. In other African countries, liberalisation of the sector expanded access and lowered prices (Gagliardone, 2016).

At the FOCAC summit in 2006, ZTE and ETC concluded a three-year sole supplier framework agreement worth $1.9 billion. Through this agreement, ZTE was given the right to supply and install all telecommunications equipment in Ethiopia over a three-year period. The award of such a large contract to one supplier, without a competitive process, was questioned by many observers, including the World Bank (Custer et al., 2023).

Later projects, such as the expansion of the network carried out in parallel by Huawei and ZTE, also faced challenges. In particular, the Ethiopian Government asked for an additional $150 million for part of the project included in the original contract. As a result, the contract was terminated in 2014, and the government approached Nokia and Ericsson, eventually reaching a deal with Ericsson. The interest rate and maturity were adjusted after negotiation between the two parties, but they both remained higher than the interest rate and maturity in the supplier credit agreement with ZTE. After ZTE learned that the government was negotiating with Ericsson, it agreed to the original plan (Custer et al., 2023).

Allegations of corruption emerged in relation to the Millennium Project. Irregularities in purchases from international suppliers and bribe-taking were raised as concerns by the Ethiopian authorities, including procurement of low-quality equipment (Ibid.).

Kenya

Benefits: In Kenya, Chinese institutions – primarily China Eximbank – have financed a combination of network expansion and e-government projects. In 2007, China Eximbank and the Government of Kenya signed a $37 million loan agreement for Phase 1 of the National Optic Fibre Backbone Infrastructure (NOFBI). The loan was provided to finance a commercial contract with ZTE and Huawei to lay 4,300km of fibre optic cable to connect 8 provincial headquarters and 53 towns. In 2012, China Eximbank and the government signed a concessional loan agreement for $72.5 million for Phase 2 of the NOFBI project and for an e-government expansion project, followed by another similar agreement in 2016, worth $98.7 million, for the same project. The funding allocated through these two loans was to be used to contract Huawei to construct 2,100km of national optic fibre backbone infrastructure, passing through 8 major towns and 36 districts. Construction began in 2014, but as of February 2021, the project was still not complete (Custer et al., 2023). In Kenya, the expansion of the network led to a huge increase in internet users: from less than 1% of the population in 2000 to over 40% in 2022. As with Ethiopia, this progress cannot be attributed exclusively to Chinese financing, but it certainly played an enabling role.

Another project that has attracted attention is the plan for Konza City (‘Africa’s first planned smart city’: Jili (2022: 44)). In 2019, the Chinese and Kenyan Governments signed a cooperation agreement to establish the Konza Data Centre, worth $665 million. China Eximbank has committed a $167 million loan for the Konza Data Centre and Smart City, which will include a national cloud data centre, smart ICT network and public safety projects (Jili, 2022). The project site is approximately 60km from Nairobi and was to be built by Huawei through an EPC contract (Custer et al., 2023), although details on the status of the project are scarce.

Challenges. Implementation of the main Chinese ICT projects in Kenya was subject to challenges and delays. The first phase of the NOFBI project faced challenges with regard to transparency. In 2022, Kenya’s Auditor General identified equipment that was faulty, idle or disconnected from electricity. It was noted that total funding for the project could not be ascertained, and that the concessional loan agreement and contract with ZTE had not been provided for audit (Custer et al., 2023).

The second phase of NOFBI was to be completed in 2016 – but remained in process as of 2025. The financing agreement with China Eximbank indicated that the government could sell excess capacity commercially to the public to finance loan repayments. However, this was not done. A parliamentary inquiry has raised concerns regarding financial management, transparency in procurement processes and the lack of revenue generation from the infrastructure (Sunday, 2021; Custer et al., 2023).

The Konza City project also experienced delays after works had commenced. In 2020, the first phase of Konza Technopolis Data Center was completed. Progress has been made on the construction of roads, a utility corridor and the installation of a water reclamation system. However, since 2021, the wider Konza City project has been stalled due to delayed government funding, corruption concerns and shifting national priorities, which have also undermined tech investors' confidence in the project (The National, 2018; Baraka, 2021). Plans for the project have evolved, and the site has received some new investment, including from Korea Eximbank, but Konza City itself has not yet provided the material benefits (jobs, incomes) that it was supposed to create (Makena, 2023).

Renewable Energy

Renewable energy projects have contributed to the expansion of energy generation capacity on the African continent, as well as providing first time access for unserved populations. Two salient cases – Ethiopia and Kenya – shed light on the development benefits and implementation challenges of this type of infrastructure.

Kenya

Benefits: In Kenya, the government has pursued an ambitious green energy agenda, with a 100% clean energy target date of 2030 and a net-zero target date of 2050. It intends to leverage a variety of renewable resources to provide universal access to energy. Kenya has attracted financing from international partners, including China, for renewable investments including in geothermal, solar and wind.

The Olkaria Geothermal field project, which comprises six planned geothermal power stations, has been financed through an array of bilateral and multilateral donors, including China Eximbank, which helped finance the $122 million drilling and exploration project in Olkaria IV, beginning in 2011. The SOE SinoPec was contracted for the pipe system installation of the 280MW power stations.

Chinese private companies have also been involved in the Menenga geothermal project, via Sosian, a Chinese-financed power producing company, which operates a 35MW geothermal plant as an independent power producer (IPP), and which began generating power for the grid in 2023 (Hakeenah, 2023).

The most prominent Chinese-financed renewable project has been the 50MW Garissa Solar Power Project, agreed in 2015. In conjunction with Kenya's Rural Electrification Authority (REA), the project was constructed by Jiangxi Corporation for International Economic and Technical Cooperation (CJIC) using Chinese materials from JinkoSolar, which supplied the solar panels. It was financed by a $155 million concessional loan from China Eximbank. The project was completed in 2019, and has since expanded its capacity to 54MW, becoming the largest solar plant in Kenya. While 5MW of its capacity serves the regional grid in Garissa, it also produces power for the national grid.

Challenges: Both geothermal and solar projects in Kenya experienced conflict with local communities. In Garissa, tensions arose over the limited employment opportunities promised by the project (Bhamidipati and Hansen, 2021). In Olkaria, issues revolved around inadequate communication and unsatisfactory resettlement policies and logistics (Kong'ani et al., 2021). Olkaria has also been scrutinised over value for money on the well-drilling investment: a report from the Auditor General in 2021 notes that some of the wells drilled between 2011 and 2015 'have never been connected to any plant for generation of power … while no corresponding revenue has been realised to date', despite the utility continuing to repay the loan.

Ethiopia

Benefits: In Ethiopia, financing from China has been instrumental in the implementation of the national Growth and Transformation Plan II, with its ambitious target of scaling up renewable energy capacity, not only from hydropower, but also by expanding wind, solar and other renewables. In addition to major hydropower projects, Chinese financing has supported the development of wind power. The development of renewable energy was tied to the country's broader infrastructure development plans, including the SGR network, which was entirely electrified.

Chinese firms and financing have supported two major wind projects: the Adama and Aysha wind farms. The 51MW Adama wind farms, south-east of Addis Ababa, were developed by the parastatal Ethiopia Electric Power (EEP or EEPCo) in partnership with HydroChina and CGCOC. This was supported as an EPC+F contract, with a preferential export credit from China Eximbank. Phase I of the project was inaugurated ahead of schedule in 2012, and Phase II came online in 2015. During construction, the wind plants generated 1,500 jobs and EEP also negotiated Chinese support for some technology transfer via capacity-building of local engineers and university academics (Chen, 2018).

China Eximbank also financed the Aysha I and II wind power projects to the value of $252 million in 2017; these were partially active as of 2023. This project was to be completed under an EPC structure with Dongfang Electric Corporation. The plan was to install two 120MW-capacity plants and was structured similarly to the Adama wind farms project, with the intention of powering the Addis–Djibouti economic corridor and to generate foreign exchange through energy exports.

Challenges: The Aysha project faced significant difficulties, being impacted by the country's macroeconomic downturn and fiscal constraints, including foreign exchange shortages. In addition, technical and land compensation issues, as well as the Covid-19 pandemic and the civil war, all contributed to substantial delays in project completion. Disbursements on the Chinese loan were halted in 2021 following Ethiopia's default and initialisation of its sovereign debt restructuring through the G20 Common Framework. Future phases of the project were shelved as of 2023.